Thomas Donoghue, PhD

Lecturer in Cognitive Neuroscience
Faculty of Biology, Medicine & Health
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Languages: English (native), French (proficient), Spanish (intermediate)

Areas of Specialization

Cognitive & Computational Neuroscience - Electrophysiology - Periodic & Aperiodic Activity

Education

2014 - 2020 **PhD, Cognitive Science - Advisor: Prof. Bradley Voytek**
University of California, San Diego, La Jolla, California, USA
Thesis: Measuring and Investigating Periodic and Aperiodic Neural Activity

2011 - 2014 **Bachelors of Arts and Sciences (BA&Sc) Honors Cognitive Science**
McGill University, Montreal, Quebec, Canada
Major: Cognitive Science. Minor: Philosophy. Graduated First Class Honors with Distinction

Professional Positions

8/2025 - *present* **Lecturer in Cognitive Neuroscience**
University of Manchester, Division of Communications, Psychology, and Human Neuroscience
Junior faculty position, including independent research, teaching, and service.

3/2021 - 8/2025 **Postdoctoral Research Scientist - Advisor: Prof. Joshua Jacobs**
Columbia University, Department of Biomedical Engineering
Investigations of human electrophysiology, with intracranial recordings and single units.

10/2020 - 2/2021 **Postdoctoral Scholar - Advisor: Prof. Bradley Voytek**
UC San Diego, Department of Cognitive Science, Cognitive & Neural Dynamics Lab
Developing software tools for the analysis of electrophysiological recordings.

9/2014 - 9/2020 **Graduate Student Researcher - Advisor: Prof. Bradley Voytek**
UC San Diego, Department of Cognitive Science, Cognitive & Neural Dynamics Lab
Mechanisms of neural communication using human electrophysiological recordings.

5/2013 - 6/2014 **Research Assistant - Advisor: Prof. Sylvain Baillet**
Montreal Neurological Institute, Department of Neurology & Neurosurgery
Functional connectivity during sleep, using magnetoencephalography and polysomnography.

9/2012 - 9/2014 **Research Assistant - Advisor: Prof. Kris Onishi**
McGill University, Department of Psychology - McGill Infant Development Cluster (MIDC)
Psycholinguistics & Developmental Psychology: language perception & statistical learning.

Preprints & Articles Currently Under Review

Underlined are students or research assistants under my direct supervision.

preprint Zhang W, **Donoghue T**, Qasim SE, Jacobs J. Evaluating Place Cell Detection Methods in Rats and Humans – Implications for Cross-Species Spatial Coding. *bioRxiv*. DOI: 10.1101/2025.08.29.672705. LINK

preprint **Donoghue T**. A historical overview of the study of aperiodic neural activity. *PsyArXiv*. DOI: 10.31234/osf.io/zrvxa. LINK

preprint **Donoghue T**, Hammonds R, Eric Lybrand, Waschke L, Gao R, & Voytek B. Evaluating and Comparing Measures of Aperiodic Neural Activity. *bioRxiv.* DOI: 10.1101/2024.09.15.613114. LINK
Project Website: https://aperiodicmethods.github.io

preprint Park SE, **Donoghue T**, Jacobs J, Lee SA. Aperiodic neural excitation of the prefrontal cortex offsets age-related decrease in hippocampal theta activity for spatial memory maintenance. DOI: 10.1101/2024.10.03.616418. LINK

preprint He W, **Donoghue T**, Sowman PF, Seymour RA, Brock J, Crain S, Voytek B, & Hillebrand A. Co-Increasing Neuronal Noise and Beta Power in the Developing Brain. *bioRxiv.* DOI: 10.1101/839258. LINK

Journal Articles (Peer Reviewed)

Underlined are students or research assistants under my direct supervision.

2026 van Engen Q, Chau G, Smith A, Adam KCS, **Donoghue T**, & Voytek B. Dissociating contributions of theta and alpha from aperiodic neural activity in human visual working memory. Journal of Neuroscience. *46*(7). DOI: 10.1523/JNEUROSCI.2340-24.2025. LINK

2025 Ameen MS, Jacobs J, Schabus M, Hoedlmoser K, **Donoghue T**. The Temporal Dynamics of Aperiodic Activity Track Changes in Sleep Architecture. Communications Psychology, *3*(160). DOI: 10.1038/s44271-025-00334-2. LINK

2025 **Donoghue T**. A systematic review of aperiodic neural activity in clinical investigations. European Journal of Neuroscience, *62*(7), e70255. DOI: 10.1111/ejn.70255. LINK

2024 Kopčanová M, Tait L, **Donoghue T**, Stothart G, Smith L, Flores Sandoval AA, Buss S, Shafi M, Pascual-Leone A, Fried PJ, Benwell CSY. Resting-state EEG signatures of Alzheimer's disease are driven by periodic but not aperiodic changes. *Neurobiology of Disease, 190*, 106380. DOI: 10.1016/j.nbd.2023.106380. LINK

2023 **Donoghue T**, Maesta-Pereira S, Han CZ, Qasim SE, Jacobs J. spiketools: a Python package for analyzing single unit neural activity. *Journal of Open Source Software, 8*(91), 5268. DOI: 10.21105/joss.05268. LINK

2023 **Donoghue T**, Cao R, Han CZ, Holman CM, Brandmeir NJ, Wang S, Jacobs J. Single neurons in the human medial temporal lobe flexibly shift representations across spatial and memory tasks. *Hippocampus, 33*(5), 600-615. DOI: 10.1002/hipo.23539. LINK

2023 Han CZ, **Donoghue T**, Cao R, Kunz L, Wang S, Jacobs J. Using multi-task experiments to test principles of hippocampal function. *Hippocampus, 33*(5), 646-657. DOI: 10.1002/hip.23540. LINK

2022 **Donoghue T,** Schaworonkow N & Voytek B. Methodological considerations for studying neural oscillations. *European Journal of Neuroscience*, *55*(11-12), 3502-3527. DOI: 10.1111/ejn.15361. LINK
Project website: oscillationmethods.github.io

2022 **Donoghue T**, Voytek B, & Ellis S. Course Materials for Data Science in Practice. *Journal of Open Source Education*, *5*(51), 121. DOI: 10.21105/jose.00121. LINK
Project website: https://datascienceinpractice.github.io/

2022 **Donoghue T** & Voytek B. Automated meta-analysis of the event-related potential (ERP) literature. *Scientific Reports*, *12*(1). DOI: 10.1038/s41598-022-05939-9. LINK
Project website: erpscanr.github.io

2022 Ostlund BD, **Donoghue T**, Anaya B, Gunther KE, Karalunas SL, Voytek B, Pérez-Edgar KE Spectral parameterization for studying neurodevelopment: How and why. *Developmental Cognitive Neuroscience*, *54*,101073. DOI: 10.1016/j.dcn.2022.101073. LINK

2021 Waschke L, **Donoghue T**, Fiedler L, Smith S, Garrett DD, Voytek B & Oblesser J. Modality-specific tracking of attention and sensory statistics in the human electrophysiological spectral exponent. *eLife*. DOI: 10.7554/eLife.70068. LINK

2021 **Donoghue T**, Voytek B, & Ellis S. Teaching Creative and Practical Data Science at Scale. *Journal of Statistics and Data Science Education*, *29*(sup1), S27-S39. DOI: 10.1080/10691898.2020.1860725. LINK

2020 **Donoghue T**, Haller M, Peterson EJ, Varma P, Sebastian P, Gao R, Noto T, Lara AH, Wallis JD, Knight RT, Shestyuk A & Voytek B. Parameterizing Neural Power Spectra into Periodic and Aperiodic Components. *Nature Neuroscience*, *23*. DOI: 10.1038/s41593-020-00744-x. LINK
Project website: specparam-tools.github.io; Media coverage: Quanta Magazine; reprinted in Wired
This paper was recognized as the 2nd most cited neuroscience paper of 2020 by The Transmitter

2020 **Donoghue T**, Dominguez J & Voytek B. Electrophysiological Band Ratio Measures Conflate Periodic and Aperiodic Activity. *eNeuro*, *7*(6). DOI: 10.1523/eneuro.0192-20.2020. LINK

2019 Robertson MM, Furlong S, Voytek B, **Donoghue T**, Boettiger CA, & Sheridan MA. EEG Power Spectral Slope Differs by ADHD Status and Stimulant Medication Exposure in Early Childhood. *Journal of Neurophysiology*, *122*(6). DOI: 10.1152/jn.00388.2019. LINK

2019 **Donoghue T**. LISC: A Python Package for Scientific Literature Collection and Analysis. *Journal of Open Source Software*, *4*(41), 1674. DOI: 10.21105/joss.01674. LINK
Project website: https://lisc-tools.github.io/

2019 Cole S, **Donoghue T**, Gao R & Voytek B. NeuroDSP: A Package for Neural Digital Signal Processing. *Journal of Open Source Software*, *4*(36), 1272. DOI: 10.21105/joss.01272. LINK
Project website: https://neurodsp-tools.github.io/

Book Chapters

2023 **Donoghue T** & Watrous A. How can we differentiate narrow-band oscillations from aperiodic activity? In *Intracranial EEG: A Guide for Cognitive Neuroscientists.* (p. 351-364) Springer, Cham. DOI: 10.1007/978-3-031-20910-9_22. LINK [Open Access Preprint: LINK]

Conference Proceedings (Peer Reviewed Papers - Selected)

Underlined are students or research assistants under my direct supervision.

2019 **Donoghue T**, Gao R, Waschke L & Voytek B. A Simulation-Based Comparison of Methods for Analyzing Aperiodic Neural Activity. *Cognitive Computational Neuroscience*. LINK

2018 Fox W, **Donoghue T**. Confidence Levels in Scientific Writing: Automated Mining of Primary Literature and Press Releases. *Proceedings of the Cognitive Science Society*. LINK

2017 Gao R, **Donoghue T** & Voytek B. Automated Generation of Cognitive Ontology via Web Text-Mining. *Proceedings of the Cognitive Science Society*. LINK

Conference Presentations

12/2022 **Investigators Workshop Presenter:** Extracting neural signals from noise.
American Epilepsy Society Meeting, Nashville, TN, USA.
Award: funding provided to attend the conference. Website: https://aessignalworkshop.github.io/

11/2022 **Short Talk:** Single neurons in the human medial temporal lobe engage in distinct aspects of different tasks. *Human Single Neuron Meeting*, Los Angeles, CA, USA.

11/2018 **NanoSymposium Presentation:** Parameterizing Neural Power Spectra
Society for Neuroscience Conference, San Diego, CA, USA.

1/2016 **Research Talk:** The Effect of Oscillatory Phase on Perception and Cognition
Temporal Dynamics of Learning Centre - All Hands Meeting, San Diego, CA, USA.

Interactive Workshops

10/2025 **specparam2.0: spectral parameterization with time-resolved estimates & update models**
Cutting EEG: Practical M/EEG (3rd Ed)., Aix-Marseille University; Toolbox Bouquet [Presented Virtually]
Developed and presented a guided demonstration of the updated spectral parameterization toolbox.

6/2023 **A practical guide to EEG analysis tools used in neuroscience of consciousness & cognition**
Association for the Scientific Study of Consciousness, New York, NY, USA.
Co-developed & presented as part of an interactive workshop on software tools for neural data analysis.

6/2023 **Advanced topics in the analysis of neural electrophysiology data: Decomposing rhythmic & broadband components**
The 36th New England Statistics Symposium, Boston, ME, USA.
Co-developed & presented as part of an interactive workshop on software tools for neural data analysis.

3/2019 **New Methods for Analyzing Periodic Oscillations and Aperiodic 1/f in Electrophysiology**
Cognitive Neuroscience Society Conference, San Francisco, CA, USA.
Developed & lead an interactive workshop covering software tools for neural data analysis.

2013 - 2015 **Brainstorm Software for M/EEG Analyses**
Assisted with interactive workshops for the Brainstorm toolbox [3 workshops].

Research Presentations (Academic - Invited Seminars)

3/2022 - **Separating periodic and aperiodic activity to investigate physiology, cognition, & disease**
BrainWorks Seminars, Wu Tsai Institute, Yale University [in person - 10/2024]
Cognitive Brown Bag, Center for Cognitive Neuroscience, Dartmouth University [in person - 3/2022]
Psychology Seminar Series, Psychology Department, University of Salzburg [virtual - 3/2022]

Research Presentations (Industry - Invited Seminars)

2018 - **Investigating Periodic & Aperiodic Neural Activity**
Friday Talk Series, Beacon Biosignals (virtual) [10/2021]
Invited Seminar, Interaxon, Toronto, Canada (in person) [8/2018]

Research Presentations (Guest Talks)

2020- **Investigating Periodic & Aperiodic Neural Activity**
University: invited talks to group meetings & journal clubs (virtual) [9 talks]

Conference Abstracts & Posters (Selected)

Underlined are students or research assistants under my direct supervision.

2025 Zhang W, Maesta Pereira S, Donoghue T, Carranza OA, Lega B, Saez I, Jacobs J. Human hippocampal neurons encoding of space and reward consistent with successor representation. *Computational and Systems Neuroscience (COSYNE)*, Montreal, Quebec, Canada.

2024 Han CZ, **Donoghue T**, Tsitsiklis M, Jacobs J. Human single neuron correlates of spatial navigation and memory performance across encoding and recall. *Society for Neuroscience*, Chicago, IL, USA.

2024 Zhang W, Maesta Pereira S, **Donoghue T**, Carranza OA, Lega B, Saez I, Jacobs J. Representations of spatial and reward structure in human hippocampal neurons. *Society for Neuroscience*, Chicago, IL, USA.

2024 Topalovic U, Azab H, Stangl M, Seeber M, Vallejo M, Batista D, Jenkens-Drake M, Hiller S, El-Gaby M, Shah S, Mathura R, Bartoli E, Watrous A, Anand A, Adkinson J, **Donoghue T**, Maesta Pereira S, Sakon J, Kurth-Nelson Z, Smith E, Inman C, Fried I, Jacobs J, Botvinick M, Behrens T, Sheth S, Suthana N. Theta representations of sequential task structure in the human medial temporal lobe during free walking. *Society for Neuroscience*, Chicago, USA.

2024 **Donoghue T**, Voytek B, & Jacobs J. specparam 2.0: spectral parameterization with time-resolved estimates and updated model forms. *FENS Forum*, Vienna, Austria. LINK
Also presented at: *Society for Neuroscience 2024,* Chicago, IL, USA

2024 **Donoghue T**, Maesta Pereira S, Qasim SE, Patel A, Azab H, Smith EH, Mathura R, Myers J, Anand A, Adkinson J, Davis TS, Kurth-Nelson Z, Rey HG, Rolston JD, Behrens TEJ, Botvinick M, Sheth SA, Jacobs J. Conjunctive Encoding in Human Place and Time Cells and Their Relation to Spatial Memory. *Interdisciplinary Navigation Symposium*, Merano, Italy. LINK
Also presented at: *Society for Neuroscience 2024,* Chicago, IL, USA

2024 Park SE, **Donoghue T**, Jacobs J, Lee SA. Can decreased hippocampal theta in older adults be restored by spatial training? *Interdisciplinary Navigation Symposium*, Merano, Italy.

2023 Ameen MS, Jacobs J, Hoedlmoser K, **Donoghue T**. The temporal dynamics of aperiodic activity track changes in sleep structure. *Society for Neuroscience*, Washington DC, USA. LINK
Also presented at: *World Sleep 2023,* Rio de Janeiro, Brazil
Also presented at: *Psychology and Brain (PUG23),* Tübingen, Germany

2023 **Donoghue T**, Maesta Pereira S, Qasim SE, Patel A, Azab H, Smith EH, Mathura R, Myers J, Anand A, Adkinson J, Davis TS, Kurth-Nelson Z, Rey HG, Rolston JD, Behrens TEJ, Botvinick M, Sheth SA, Jacobs J. Conjunctive encoding in human place and time cells and their relation to spatial memory. *Society for Neuroscience*, Washington DC, USA. LINK

2023 Azab H, El-Gaby M, Shah S, Mathura R, Bartoli E, Watrous A, Anand A, Atkinson J, **Donoghue T**, Maesta Perreira M, Topalovic U, Sakon J, Kurth-Nelson Z, Smith E, Suthana N, Fried I, Jacobs, J, Botvinick M, Behrens T, Sheth SA. Single neuron representations of sequential task structure emerge rapidly in human anterior cingulate and entorhinal cortex. *Society for Neuroscience*, Washington DC, USA.

2023 Park S, **Donoghue T**, Jacobs J, Lee SA. Aperiodic and periodic intracranial EEG correlates of aging & age-related spatial memory decline. *Society for Neuroscience*, Washington DC, USA.

2023 Han CZ, **Donoghue T**, Kunz L, Jacobs J. Human single neuron correlates of spatial navigation and memory performance. *Society for Neuroscience*, Washington DC, USA.

2023 Zhang W, **Donoghue T**, Qasim SE, Jacobs J. Variability across methods in the identification & characterization of place cells in humans data. *Society for Neuroscience*, Washington DC, USA.

2022 **Donoghue T**, Kleen JK, Voytek B, Jacobs J. Methodological considerations for examining spectral features in epilepsy. *American Epilepsy Society Meeting*, Nashville, TN, USA. LINK

2022 Maesta Pereira S, **Donoghue T**, Qasim SE, Patel A, Azab H, Smith EH, Mathura R, Myers J, Anand A, Adkinson J, Davis TS, Shofty B, Kurth-Nelson Z, Rey HG, Rolston JD, Behrens TEJ, Botvinick M, Sheth SA, Jacobs J. Conjunctive encoding in human place and time cells. *Human Single Neuron Meeting,* Los Angeles, CA, USA. LINK
Also presented at: *Society for Neuroscience,* San Diego, CA, USA.

2022 **Donoghue T**, Cao R, Han CZ, Holman CM, Brandmeir NJ, Wang S, Jacobs J. Single neurons in the human medial temporal lobe engage in distinct aspects of different tasks. *Human Single Neuron Meeting,* Los Angeles, CA, USA. LINK
Also presented at: *Society for Neuroscience,* San Diego, CA, USA.

2021 **Donoghue T**, Qasim SE, Patel A, Azab H, Smith EH, Mathura R, Myers J, Anand A, Atkinson J, Rey HG, Rolston JD, Behrens TEJ, Botvinich M, Sheth SA, Jacobs J. Human single neuron activity encodes future trajectories. *Society for Neuroscience*, Virtual.

2020 **Donoghue T** & Voytek B. Considerations for Detecting & Measuring Neural Oscillations. *LiveM/EEG (Cutting EEG)*, Virtual Conference. LINK

2019 Farnan T, **Donoghue T**, Voytek B. Evaluating Spectral Estimation Methods for Time-Resolved Measurement of Aperiodic Activity. *Society for Neuroscience*, Chicago, IL, USA. LINK

2019 Zhang F, **Donoghue T**, Voytek B. Comparing the Effects of Pre-Stimulus Periodic and Aperiodic Activity on Post-Stimulus Event Related Potentials. *Society for Neuroscience*, Chicago, IL, USA. LINK

2019 Waschke L, **Donoghue T**, Smith S, Voytek B, & Obleser J. Tracking of 1/f Stimulus Characteristics in the Human EEG. *Society for Neuroscience*, Chicago, IL, USA.

2019 **Donoghue T**, Gao R, Waschke L, Voytek B. A Simulation-Based Comparison of Methods for Analyzing Aperiodic Neural Activity. *Cognitive Computational Neuroscience,* Berlin, Germany. LINK

2019 Dominguez J, **Donoghue T**, Voytek B. Electrophysiological Frequency Band-Ratio Measures Conflate Changes in Periodic and Aperiodic Features. *Cognitive Neuroscience Society*, San Francisco, CA, USA. LINK

2018 Mdanda L, **Donoghue T**, Voytek B. Parameterization of Periodic and Aperiodic Human Electrophysiology Reveals Greater Between- Than Within-Subject Variability. *Society for Neuroscience*, San Diego, CA, USA. LINK

2018 **Donoghue T**, Sebastian P, Voytek B. Topographical Analysis of Electrophysiological 1/f and Oscillations Reveals Patterns of Spatial Variation. *Biomag,* Philadelphia, PA , USA. LINK

2018 **Donoghue T**, Sebastian P, Noto T, Haxby S, Voytek B. Integrating Human Electrophysiology, Gene Expression and Functional Data. *Neuroinformatics*, Montreal, QC, Canada. LINK

2018 Fox W, **Donoghue T**. Confidence Levels in Scientific Writing: Automated Mining of Primary Literature and Press Releases. *Cognitive Science,* Madison, WI, USA. LINK

2018 **Donoghue T** & Voytek B. Alpha Power and 1/f Slope Provide Independent Decoding of Visual Spatial Attention. *Cognitive Neuroscience Society*, Boston, MA, USA. LINK

2018 Gao R, **Donoghue T,** Voytek B. Defining Cognition: Automated Generation of Cognitive Ontology by Text-Mining Literature. *Cognitive Neuroscience Society*, Boston, MA, USA.

2017 Waschke L, **Donoghue T,** Obleser J, Voytek B. Attention-Modulated Tracking of 1/f Stimulus Characteristics in Human EEG. *Signals & Noise in the Auditory Pathway*, Lübeck, Germany.

2017 **Donoghue T** & Voytek B. Assessing approaches for estimating the electrophysiological 1/f background spectrum. *Society for Neuroscience*, Washington DC, USA. LINK

2017 **Donoghue T** & Voytek B. Automated meta-analysis of event-related potentials and their correlates by text-mining. *Cognitive Neuroscience Society*, San Francisco, CA, USA. LINK
Award: graduate student award winning poster including a 500$ travel award

2016 **Donoghue T**, Fox W, Kim A, Voytek B. The relation of oscillatory-phase to visual perception depends on attention & location of stimuli. *Society for Neuroscience*, San Diego, CA. LINK

2016 Sebastian P, **Donoghue T**, Noto T, Haxby S, Voytek B. Data mining to generate novel hypotheses for the genetic underpinnings and functional roles of cortical oscillations. *Society for Neuroscience*, San Diego, CA, USA. LINK

2016 **Donoghue T**, Sebastian P, Voytek B. Automated Analysis of Resting State Cortical Oscillatory Characteristics using Magnetoencephalography. *Biomag,* Seoul, South Korea. LINK

2015 Gougelet R, **Donoghue T,** Piper M, Althoff A, Urbach TP, Voytek B. Influencing Visual Target Detection with Oscillatory Phase-Specific Stimulus Presentation. *Society for Neuroscience*, Chicago, IL, USA. LINK

Honors & Awards

2024 **Travel Award, FENS Forum**
Merit based, competitive application, award for FENS forum 2024 travel support ($750 Euros).

2023 **The Neuro Irv & Helga Cooper Foundation Open Science Prizes, Trainee Prize (Runner Up)**
Award for work on Open Science practices & tools with an impact in neuroscience ($1000 CDN).

2022 **Trainee Professional Development Award, Society for Neuroscience (SfN)**
Merit based award for SfN 2022 conference registration and travel funds ($1000 USD).

2017 - 2019 **Travel Awards, UC San Diego, Graduate Student Association (GSA)**
Travel awards for conferences, from the GSA (2X) and from departmental funds.

1/2016 **Small Grants Award, Temporal Dynamics of Learning Centre (TDLC)**
Funding for an EEG project on the temporal dynamics of perceptual learning ($2200 USD).

3/2014 **Owens Scholar Award, Johns Hopkins University** (*declined*)
Additional funding over 3 years offered with admission to Johns Hopkins ($18 000 USD).

11/2013 **Samuel de Champlain Quebec Program for International Collaboration**
Funds provided by my research supervisor (Dr. Baillet) for travel to NeuroSpin in France.

Academic Activities: Reviewing

**Includes article co-reviewed with a research supervisor. #Includes code review.*

Journal Articles (Ad-Hoc Reviewer)

The American Journal of Psychiatry (1X); Behavior Research Methods (1X); Biological Psychology (1X); Brain and Behavior (1X); Cerebral Cortex (2X); Clinical Neurophysiology (1X); Cognitive Neurodynamics (1X); Developmental Cognitive Neuroscience (3X); Developmental Psychology (1X); eLife (2X); eNeuro (2X); European Archives of Psychiatry and Clinical Neuroscience (1X); European Journal of Neuroscience (3X); F1000 Research (1X); GeroScience (1X); Human Brain Mapping (*1X); Imaging Neuroscience (5X); Journal of Clinical and Experimental Neuropsychology (1X); Journal of Cognitive Neuroscience (1X); Journal of Neurophysiology (*2X); Journal of Neural Engineering (1X); Journal of Neuroscience (9X); Journal of Neuroscience Methods (1X); Journal of Open Source Education (#3X); Journal of Open Source Software (#4X); Journal of Physiology (1X); Mindfulness (1X); Nature Communications (1X); Neurobiology of Aging (*2X); Neurobiology of Disease (1X); NeuroImage (5X); Open Journal of Signal Processing (1X); PLoS Computational Biology (3X); PLoS Biology (1X); Proceedings of the National Academy of Sciences (PNAS; 1X); Psychophysiology (2X); Psychonomic Bulletin & Review (1X); ReScienceC (#1X); SoftwareX (1X); Social, Cognitive, and Affective Neuroscience (2X);

Conference Proceedings

Affective Computing & Intelligent Interaction (ACII 2019: 1 paper); Cognitive Computational Neuroscience (CCN 2019: 6 papers);

Books

Columbia Press (1X);

Research Supervision & Mentorship

Students under my supervision (official) and/or for whom I serve as a research mentor (unofficial). Awards listed are those for whichI supervised the application and project.

PhD Student Supervision [University of Manchester]:

Cliodhna Hall	*incoming*	MRC studentship
Owen Page	*incoming*	Bicentennial studentship

Masters Student Supervisor [University of Manchester]:

Chloe Hooper	10/2025 - *current*

PhD Student Mentorship [Columbia University]:

Mohamed Ameen	11/2022 - 10/2025
Weija Zhang	09/2022 - *current*
Sandra Maesta Pereira	09/2021 - *current*
Zhixian (Claire) Han	09/2021 - *current*

Masters Student Research Assistant Mentorship [UC San Diego]:

Tyler Farnan	01/2019 - 03/2021

Undergraduate Research Assistants Mentorship [UC San Diego]:

Fenglin (Allen) Zhang	01/2019 - 03/2021	
Julio Dominguez	06/2018 - 01/2020	TRELS Scholarship
Luyanda Mdanda	10/2016 - 01/2020	HDSI Undergrad Fellowship
Meyhaa Buvanesh	04/2019 - 06/2019	
Lakshmi Menon	04/2019 - 09/2019	
Fiona Cisternas	01/2019 - 06/2019	HDSI Undergrad Fellowship
Priyadarshini Sebastian	10/2015 - 06/2018	FISP Trainee Award
Aeri Kim	10/2015 - 12/2016	
Will Fox	06/2015 - 06/2018	

Computational Skills & Contributions

Languages Fluent in **Python**, **shell** scripting (bash) & **git**, intermediate in **Matlab** and **R**.

Github Code & open-source contributions are available on my Github profile and indexed here.
Example contributed projects: spikeinterface, pynwb, nwbwidgets.

Open-Source Package Development & Contributions

The following are open source Python packages which I (co-) develop and maintain.

specparam (formerly 'fooof'): Spectral Parameterization (Github - PYPI - Documentation)
Lead Developer - Python package for parameterizing neural power spectra.

neurodsp: Neuro Digital Signal Processing (Github - PYPI - Documentation)
Co-Developer - Python package for analyze neural electrophysiological recordings.

lisc: Literature Scanner (Github - PYPI - Documentation)
Lead Developer **-** Python package for collecting and analyzing the scientific literature.

spiketools: Analysis of spiking data (Github - PYPI - Documentation)
Lead Developer - Python package for analyzing single-unit neural data.

ByCycle: Cycle-by-cycle analysis of neural oscillations (Github - PYPI - Documentation)
Maintainer - A package for analyzing cycle properties of neural oscillations.

Pipelines & Template Contributions

The following are openly available pipelines and templates I have created and/or work on.

HSNPipeline: Human Single-Neuron Pipeline (Github - Documentation)
Lead Developer - Pipeline with templates & resources for processing human single-neuron data.

StructuredScience (Github - Website)
Lead Developer - Curated templates and resources for creating organized scientific projects.
This resource curates templates for creating organized and standardized project structures.

Resource Contributions

The following are open source / open access resources that I have created and made public.

SigViz: animated signal visualizers for exploring signal processing (Github - Website)
This resources includes novel animations to explain signals and topics such as filtering.

OpenLists: open lists of open resources (Github - Website)
This collection curates open resources, including open-access data & open-source tools.

Additional Training in Computational Skills

The following are additional short courses I have completed related to developing computational skills.

2018 **Methods In Neuroscience at Dartmouth (MIND)**, *Dartmouth College*, Hanover, NH, USA
Short course.Topic: Narratives & Natural Contexts. Competitive application (~20% acceptance).

2017 **Neurohackweek,** *eScience Institute*, University of Washington, Seattle, WA, USA
Project-based course on neuro- & data science. Competitive application (~25% acceptance).

2016 **Advanced Scientific Programming in Python**, *G-Node & CINN*, Reading, England, UK
Short course on scientific programming. Competitive application (9.9% acceptance).

Training in Teaching

Formal training in teaching and related topics.

2018 **Introduction to College Teaching**, Teaching & Learning Commons, UC San Diego
Semester long course on evidence-based teaching in university contexts.

2017 **Equity, Diversity, and Inclusion in Postsecondary Education**, UC San Diego Extension
Course on best practices for inclusive & equitable teaching in university contexts (10 hrs).

Teaching Experience

2018 **Instructor-of-Record**, Department of Cognitive Science, UC San Diego
COGS 18: Introduction to Python (30 hours lecture + coding labs; 200 undergrad students)
Developed & taught a course teaching introductory Python programming. Materials: LINK

2017 - 2020 **Instructor (3X)**, Clubes de Ciencia Mexico
Clubes de Ciencia is a non-profit organization promoting science education across Mexico.
1 week, hands-on research focused courses (25 hours of instruction; 12-18 students / year)
- *CdeCMx Challenge: Soluciones cientificas a problemas emergentes* (online, Aug. 2020)
- *Inteligencia Biologica & Artificial: Amigos o Enemigos?* (Ensenada, Mexico, Aug. 2019)
- *Bots on the Brain: Cognitive Science & Bio-Inspired Robotics* (Monterrey, Mexico, Aug. 2017)

2015 - 2017 **Instructor (3X)**, Academic Connections, UC San Diego
Academic Connections offers university-level courses to advanced high school students.
Co-developed & taught a course introducing cognitive science. Materials: LINK
Introduction to Cognitive Science (75 hours of instruction; 16-24 students / year)
Ratings: Course {4.71, 4.80, 4.59}/5; Instructor: {4.86, 4.92, 4.92}/5; Years: {2015, 2016, 2017}.

2015 - 2018 **Teaching Assistant (7X),** Department of Cognitive Science, UC San Diego
COGS 108: Data Science in Practice (Winter '18, Prof. Bradley Voytek, TA Evals: 4.31/5)
COGS 108: Data Science in Practice (Spring '17, Prof. Bradley Voytek, TA Evals: 4.32/5)
COGS 107B: Systems Neuroscience (Winter '17, Prof. Douglas Nitz, TA Evals: 4.60/5)
COGS 17: Neurobiology of Cognition (Winter '16, Dr. Christine Johnson, TA Evals: 4.58/5)
COGS 9: Introduction to Data Science (Fall '15, Prof. Bradley Voytek, TA Evals: 4.34/5)
COGS 3: Introduction to Computing (Spring '15, Prof. Bradley Voytek, TA Evals: 4.54/5)
Awarded *Excellence in Teaching* Award from the UCSD Cognitive Science Dept.
COGS 107B: Systems Neuroscience (Winter '15: Prof. Douglas Nitz, TA Evals: 4.69/5)
Awarded *Outstanding Teaching* Award from the UCSD Cognitive Science Dept.

Educational Materials

Openly available educational materials that I have created and/or contributed to.

Introductory Python, openly available online course
Materials for learning introductory programming in Python (Website - Source).

Data Science in Practice, openly available online course
Materials for learning introductory data science in Python (Website - Source).

Tutorials, openly available tutorial materials online course
Python Boot Camp: Open materials for a graduate student bootcamp. LINK
Electrophysiology Tutorials: Materials for getting started with M/EEG analyses. LINK

Science Outreach

1/2022 - 9/2023 **Mentoring: Student Mentor for High-School Summer Internship**
Mentor for the Brainyac program, offering intensive summer internships to young students.

2020 - **Mentoring: Project Guidance & Assistance with Grad School Applications**
Organizations include: Cientifico Latino, neuromatch

2018 **Public Workshops & Presentations**
10/2018: Data Wrangling & Web Scraping: 2 hr workshop with SCALE-SD. *Materials* - *Media*

2013 - 2020 **Volunteer Tutoring & School Presenter**
Tutoring, presentations, science fair judging, and miscellaneous volunteering.
Organizations include: Brain Awareness, San Diego Science Fair, San Diego Refugee Tutoring

1/2014 - 1/2017 **Science Writer / Editor / Podcast Host, Useful Science Organization** (usefulscience.org)
Writing clear, concise and useful summaries of scientific research for a general audience.